

Course Code: EST100

Course Name: ENGINEERING MECHANICS  
(2019-Scheme)

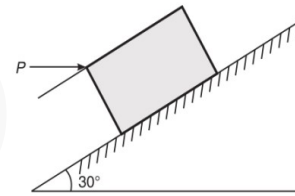
Max. Marks: 100

Duration: 3 Hours

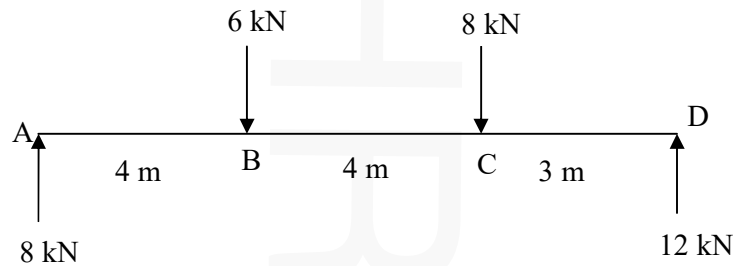
## PART A

(Answer all questions, each carries 3 marks.)

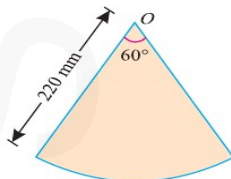
- 1 State and explain Lami's theorem. (3)
- 2 What is meant by Free body diagram? Explain with an example. (3)
- 3 A small block of weight 1000 N as shown in Figure, is placed on a  $30^\circ$  inclined plane with  $\mu = 0.25$ . Determine the horizontal force to be applied for impending motion down the plane (3)



- 4 A rigid bar AD is acted upon by forces as shown in figure below. Reduce the force system to a single force- system and locate the point of application of the single force. (3)



- 5 Find the moment about C(-2,3,5) of the force  $F = 4\hat{i} + 4\hat{j} - 1\hat{k}$  passing through the point A (1,-2,4). (3)
- 6 Find the centre of gravity of lamina from O. (3)



- 7 A 50 kg mass has a velocity of 10m/s horizontally on a smooth surface. Determine the magnitude of horizontal force required to bring the mass to rest in 5 seconds. (3)

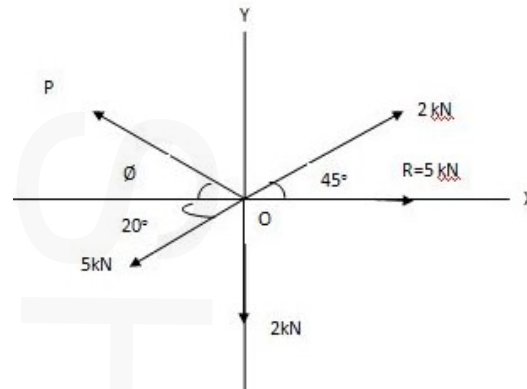
- 8 A body is projected at an angle such that its horizontal displacement is 3 times that of maximum height. Find the angle of projection. (3)
- 9 A motor car is uniformly accelerated from 40 kmph to 50 kmph over a distance of 300m. If the wheels are 1 m diameter, find the angular acceleration of wheels. (3)
- 10 Differentiate between curvilinear motion and projectile motion. (3)

**PART B**

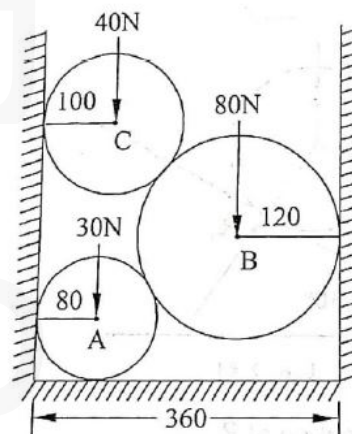
*(Answer one full question from each module, each question carries 14 marks)*

**Module-I**

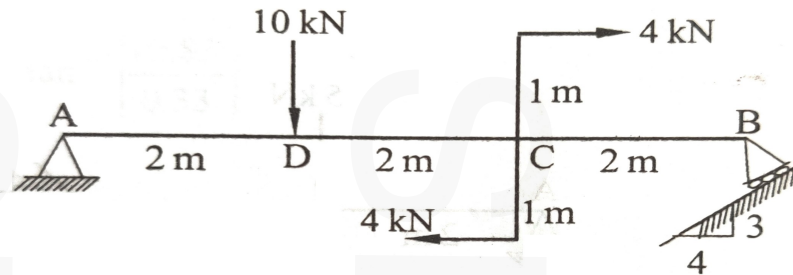
- 11 a) A rope 9m long is connected at A and B, two points on the same level, 8m apart. A load of 300N is suspended from a point C on the rope 3m from A. What load connected to point D, on the rope, 2m from B is necessary to keep portion CD parallel to AB. (5)
- b) The resultant of a system of four forces is 5kN directed towards right along X-axis. Find the force P and its direction  $\theta$ . (9)



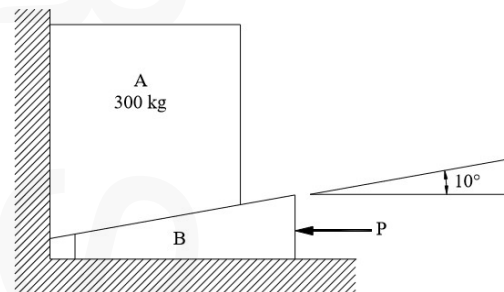
- 12 Three cylinders are piled in a rectangular ditch as in Fig. Neglecting friction, determine the reaction between cylinder A and vertical wall. (14)

**Module-II**

- 13 a) A beam 6 m long is loaded as shown. Calculate the reactions at A and B. (7)

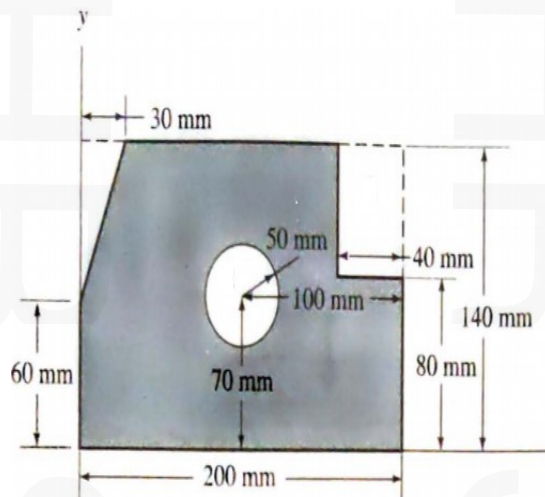


- b) The uniform ladder is of mass 10 kg and 2-m long, leaning against a vertical wall. The coefficient of static friction at A (wall) is 0.6 and at B (floor) is 0.4. Determine the smallest angle, for which the ladder can remain in the equilibrium. (7)
- 14 If the coefficient of static friction equals 0.3 for all surfaces of contact, determine the smallest value of force  $P$  necessary to raise the block A of mass 300 kg. Neglect the weight of the wedge B. Angle of wedge is  $10^\circ$ . (14)

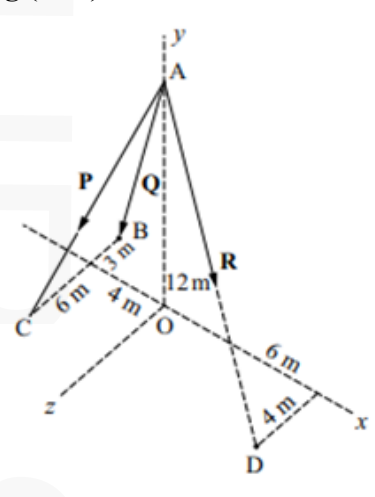


### Module-III

- 15 Find the centroid of the shaded area shown. **Fig (Q15)** (14)



**Fig (Q15)**



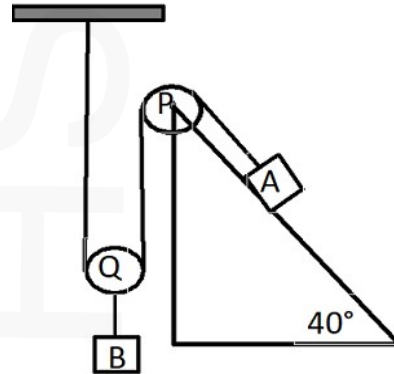
**Fig (Q16)**

- 16 Find the resultant of the force system shown in Fig. in which  $P = 280$  N,  $Q = 260$  N and  $R = 210$  N. **Fig (Q16)** (14)

### Module-IV

- 17 Determine the tension in the inextensible string and the acceleration of the (14)

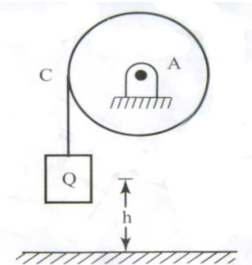
masses. Consider the pulley as massless and coefficient of friction as 0.20. Block A=200 kg and block B=100kg



- 18 a) A glass ball is dropped on to a smooth horizontal floor from which it bounces to a height of 9m. On the second bounce, it rises to a height of 6m. From what height the ball was dropped and what is the coefficient of restitution between the glass and the floor? (5)
- b) Two cars A and B travelling in same direction get stopped at a traffic signal. When signal turns green, car A accelerates at  $0.75 \text{ m/s}^2$  and 1.75 seconds later, car B starts and accelerates at  $1.1 \text{ m/s}^2$ . Determine i) when and where B will overtake A and ii) the speed of each car at that time. (9)

#### Module-V

- 19 A circular disc of radius  $r=30\text{cm}$  and weight  $W=145\text{N}$  is free to rotate about its geometric axis. A flexible cord carrying a weight of  $Q=45\text{N}$ , is wound around the circumference of the disc as shown in Fig. If the weight Q is released from rest, find (a) the time  $t$  required for it to fall through the height  $h=300\text{cm}$  (b) with what velocity  $v$  will it strike the floor? (14)



- 20 a) A 50N weight is suspended from a spring of constant  $k=8 \text{ N/cm}$ . Neglecting the mass of the spring, find the period for small amplitudes of vertical oscillations. (5)
- b) A particle performing Simple harmonic motion. When it is at distances of 10.0cm and 20.0cm from the mean position, its velocities are 1.2 m/s and 0.8 m/s respectively. Find (a) amplitude of oscillations. (b) time period of oscillations (c) its maximum velocity and (d) its maximum acceleration. (9)

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